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MindSight

Project Report

[GitHub](#)

[Website](#)

Motivation & Introduction:

In today's fast paced world, mental health and their challenges are starting to become very common. Stress, anxiety, and emotional fatigue are not just unique issues, as they are everyday experiences for millions of people. Despite it being such a crucial issue, many people still struggle to understand what truly influences their stress, mood, and well-being. Although there are multiple wellness or journaling apps available, most of the apps only focus on data collection, as they allow the user to log their mood, track habits, or even reflect on their daily activities. Aside from these tasks, users are just with a long list of entries that they created, mood logs, or just some statistics that lack any meaningful interpretation. Without personalized insights, correlations, and patterns, people might miss important signals that could help them improve their mental health or their mood. We believe its lifestyle factors such as screen time, caffeine intake, sleep quality, and exercise habits play a significant role on a person's mental health, but they are oftentimes subtle and overlooked. Some individuals even turn these factors into a habit, and aren't able to visualize how this habit is impacting their mood and also heightening their stress levels. This is where we believe Mindsight comes in, as our website will take your daily journals and turn them into meaningful analysis that will help the users see patterns and correlation between their stress level, habits, and overall well-being.

System Functionality and Approach

MindSight uses artificial intelligence and machine learning to transform raw user inputs into actionable insights. The platform analyzes daily data to predict stress levels, identify emotional patterns, and detect potential triggers that may negatively affect mental health.

MindSight provides a daily check-in for users to share lifestyle information, including screen time, caffeine intake, and physical activity. Users can also submit journal-style text entries to express their thoughts. By combining numerical and textual data, MindSight supports both statistical modeling and emotional analysis when delivering insights.

Tech Stack

In terms of the technology stack, we wanted to create a modern full-stack web application that is responsive, scalable, and allows for quick iterations. This resulted in us choosing the project's front end to use both Next.js and React, which would allow for a quick component-based user interface with responsive routing. The web pages' UI/UX is designed using Tailwind CSS, Radix UI, and FramerMotion. The designs were replicated from the Figma designs we initially designed when brainstorming for our frontend. We used built-in libraries in React called Recharts to create the stress trend and emotion distribution charts with user interactions.

The next technology stack we ended up choosing was for the database, as whichever database we choose would serve as the project's main backend platform for data storage and authentication. We ended up choosing Firebase, and we used it for storing user profiles, daily check-ins, stress predictions, and identifying emotions. Firebase was also used for a second reason, and it was for authentication. We implemented Firebase to manage secure user sign-up, login, and email verification. Each user can keep their own separate collection of entries thanks to Firestore's hierarchical data model, which guarantees data privacy and scalability.

The project also incorporates a lightweight Python FastAPI backend to enable machine learning functionality. This service uses NLP models trained and assessed in Google Colab to handle emotion detection and stress-related inference. In order to improve the interpretability and utility of the insights, we send the responses from the model to the Gemini API. This is then utilized to summarize the results and provide users with tailored recommendations based on their past behavior from their surveys.

Methods

In the development of MindSight, we used Random Forest and NLP emotion detection models, LLM, and full-stack web development technologies. Our system uses structured numeric data and unstructured text input from the user to analyze and provide individualized insights.

Data Gathering and User Input

1. Through a daily check-in (survey) system, the application gathers two primary types of data:

structured lifestyle information, such as:

- Screen time
- Caffeine intake
- Exercise frequency
- Sleep-related indicators

2. Unstructured textual data, where users can write an open-ended response in natural language to express their feelings, and experiences specific to the day.

MindSight can do both statistical prediction and emotional text analysis thanks to this dual-input method.

NLP Model for Mood and Emotion Detection

We developed an NLP-based emotion detection system to analyze the user journal entry. Custom models for sentiment and emotion classification were trained in early experiments. However, due to the limited hardware resources, training NLP models from scratch turned out to be very time-consuming. Also, deploying such a model would require significant storage on the server.

We switched to a BERT-based (Bidirectional Encoder Representations from Transformers) model to solve this issue. Google Colab, which gave access to the GPUs required for transformer-based architectures, was used to train and assess the model. We ended up choosing the BERT-base model because it excels at understanding meanings in text, it can get long-range dependencies in languages and also because it performs well specifically for emotional and sentimental-based tasks. How the model works is that the user enters information into our survey, and the information is sent to our backend, which then predicts the emotional states of the user such as positivity, negativity, stress-related sentiment, or emotional tone.

Model Training Environment

All NLP model training that we conducted was performed on Google Colab because of its GPU support and easy cloud based access. Training the transformer-based models locally was not an easy task to do because of our hardware limitations. Google Colab helped with fine-tuning pre-trained BERT models, experimenting with different hyperparameters, and lastly with evaluating performances efficiently. Despite this, the training still required a long time due to the size of the dataset we had and the complexity of the model.

Frontend Interface and User Experience

MindSight's home page greets users with a visually appealing hero section with call-to-action buttons, a features section that explains how our app works, and a mission section that talks about our motivation and vision behind this project. On our survey page, logged in users can complete daily check-ins, and see stress/emotional trends on our user-friendly interactive dashboard. To guarantee a consistent visual design and seamless user interactions across various devices, we used Tailwind CSS, and Radix UI.

Recharts were used to create our data visualizations, such as weekly stress and mood trends, and emotion distributions, which made it easy for users to understand how their emotional and behavioral patterns changed over time. To maintain application state and guarantee real-time updates as new data becomes available, React hooks were heavily utilized.

Authentication and User Data Management

Users were authenticated and data was stored using Firebase Authentication and Cloud Firestore. By providing secure sign-up and login features, including email verification, Firebase Authentication guarantees that user accounts are safe and uniquely identified. Each authenticated user is assigned a unique identification (UID) so that we can store their data in unique Firestore collections. User-specific document paths are used to store timestamps, recognized emotions, predicted stress levels, and daily check-ins. This structure ensures data privacy, and efficient querying for our dashboard.

Data Pipeline and Storage Workflow

Following a user's daily check-in, the frontend transmits journal text and structured lifestyle data to backend machine learning services. Deployed models detect emotions and

predict stress, and the frontend receives the results. As part of the user's historical record, these outcomes are subsequently saved in Firestore along with the initial inputs. The seamless integration of data collection, machine learning inference, persistent storage, and visualization is made possible by this end-to-end pipeline. The dashboard displays weekly trends, emotion distributions, and summary insights that are created by retrieving stored data.

Challenges Faced

- 1. Training Time and Computational Cost.** One of the major challenges we faced was the long training time required for NLP models. Transformer-based models such as BERT are computationally intensive, even when fine-tuning rather than training from scratch. Training sessions often took several hours, and multiple runs were needed to adjust parameters (learning rate, batch size, number of epochs).

This significantly slowed experimentation and model iteration.

- 2. Model Accuracy Limitations:** Reaching a high classification accuracy was another difficulty. The maximum accuracy attained during training was roughly 76%, despite a great deal of tuning and experimentation. This performance demonstrated the challenge of modeling subjective emotional language, particularly when user journal entries differ greatly in tone, length, and structure, even though it is acceptable for emotion detection tasks. The low accuracy highlighted how crucial it is to select a robust pre-trained model instead of relying solely on a custom-trained solution.
- 3. Dataset Complexity and Subjectivity** Emotion and mood classification is inherently subjective. Different users can express their emotions in different ways, and the same words can be said in a different tone, which would convey different emotional meanings

depending on context. This is what made our labeling and prediction for some of these emotions much more difficult compared to traditional classification tasks. Additionally, we had to factor in if the users put in short or vague journal entries that would lack enough context for the model to be able to provide a confident emotion detection.

Model Selection Justification

During the development of our app, we initially tried to train our own NLP-based emotion classification model using Google Colab and some dataset we found on kaggle. However, despite doing extensive experimentation and tuning the parameters, the custom-trained model only managed to reach a maximum accuracy of 76% and had a fewer range of emotions to choose from. This result demonstrated that the model we used was able to learn meaningful emotional patterns, but to actually further improve the model would require longer training times and additional resources.

To resolve our issue of reliability and performance in a real user-facing environment, we choose a pretrained BERT-based transformer model which we found from huggingface. This model ended up outperforming our custom-trained model. The BERT (new NLP model) demonstrated stronger understanding of emotional language and better generalization across different writing styles, and was easier to deploy as well.

By leveraging a pre-trained transformer-based model rather than relying solely on a custom-trained solution, we were able to balance performance, reliability, and development feasibility. As a result, BERT was integrated into the final application to deliver more consistent and trustworthy emotion detection.

Conclusion

MindSight shows how we can use AI and machine learning in a practical way that supports emotional wellness. Existing stress/mood tracker applications collect a lot of data from the user. But they fail to identify and show correlation between various lifestyle factors and stress/mood. With our application, we don't simply just collect user data and track their mood or stress level. Our system brings together everyday lifestyle data like sleep hours, screen time, physical activity, etc. along with an open-ended journal entry to turn users' daily checkins into something meaningful, and provides actionable emotional insights to our users. This way our app helps users better understand not just how they feel, but why they might be feeling that way.

While working on MindSight, we ran into a few challenges, especially with the emotion detection part using NLP. Training our own model to detect the user's emotion based on text entry proved difficult because we were using Google Colab, and had access to limited resources. Every time we ran our model, it took a long time to train. Because of this, we had to shift to a pretrained model instead. This offered decent results while still being realistic to implement and deploy.

Besides the machine learning models themselves, a major focus of our project was creating a smooth, user-friendly interface. We were able to accomplish that using Next.js, Tailwind CSS, and various React libraries. It was also essential to present insights through clear visualizations on a simple yet highly effective dashboard. These design choices ensure that our users can easily reflect on their emotional and stress patterns without feeling overwhelmed.

MindSight is not meant to diagnose or replace professional mental health care. Instead, it is a tool for self-awareness and reflection. By helping users notice patterns, habits and possible stressors over time, it encourages more mindful lifestyle choices and better understanding of emotional wellness. Looking ahead, future improvements could include a more dynamic dashboard with real-time feedback/suggestions based on user's recent checkins, personalized model fine-tuning, larger datasets, and stronger long-term analysis. Overall, MindSight meets its goal of using AI and machine learning to deliver meaningful and personalized insights that empower users to better understand themselves.